BL-23 and BL-24 - Low-attendance notifications and the below-threshold list

Backlog items: BL-23 (#23, FR-14) and BL-24 (#24, FR-13) | Use case UC-10 | Rules BL-01 D-11, D-13, D-14, D-15, D-03.

BL-23 - Student notification

Acceptance criterion	How it is met
Notification through the agreed channel	In-app notification (**Notifications** in the navigation bar, `/notifications/`) and an email to the student's account address (D-14).
Identifies the shortage clearly	The message lists each subject and/or the overall figure that fell below, as `attended/held (percentage)`, and the threshold.
Frequency without unintended duplicates	`ShortageFlag` stores what is currently below for each student (one row per subject, one for overall). A notification is created only for a flag that did not exist. Several percentages dropping in one save produce one notification. Students are row-locked during evaluation, so concurrent saves cannot notify twice.
Boundary rule	`attended x 100 < threshold x held`; equal is not below; no classes is never flagged (D-11, D-13).
Stops after recovery	Recovering to >= threshold deletes the flag and sends nothing. A later drop notifies again (UC-10 3a).
Only the affected student	The notification and email go to that student's account only. The inbox shows only the signed-in student's notifications; staff get 403 (permission `notifications.view`, Students only).

When the check runs: after every attendance submission, every correction, and every threshold change (which re-evaluates all active students). It runs inside the same database transaction as the change. Emails are sent after the transaction commits, so a rejected save sends nothing. If the email fails (or the account has no email address), the in-app notification is kept and an audit entry `notification.email_failed` is written (D-14, UC-10 4b). Deactivated students are not evaluated (D-03).

BL-24 - Below-threshold list for Admin and Faculty

Acceptance criterion	How it is met
Students below the configured/default threshold	`/attendance/shortages/` (**Students below threshold** on the dashboards), using the same exact comparison.
Identifies the student and percentage or subject	Roll number, name, class/section, subject and `attended/held (percentage)`. The Admin also sees overall percentages.
Changes and corrections refresh the list	Calculated from saved records on every request; nothing is cached.
Recovered students leave the list	Only students currently below are listed.
Authorized scope	Permission `low_attendance.view` (Admin, Faculty). Faculty see only their assigned subjects, and only classes they teach are counted. Overall rows are Admin-only, because an overall figure includes other faculty members' classes. Deactivated students are left out.
Empty list message	"No students are currently below the 75% threshold ...".
Monday 08:00 email; none when empty	`python manage.py send_weekly_shortage_emails`. Each active faculty member with a non-empty list gets one email of their own students. At most one per faculty member per week (`WeeklyShortageEmail`), so re-running is safe; a failed send is logged (`notification.weekly_email_failed`) and retried on the next run.

Scheduling the Monday email

The project has no background worker, so the server's cron runs the command:

```
CRON_TZ=Asia/Kolkata
0 8 * * 1  cd /path/to/Student-Attendance-Management-System && .venv/bin/python manage.py send_weekly_shortage_emails
```

Email delivery uses the existing `EMAIL_*` settings (printed to the terminal by default).

Tests

`python manage.py test apps.notifications` - covers UT*10*1 to UT*10*8, IT*10*4 to IT*10*6, ST*10*1 to ST*10*6, scope, privacy, empty list, weekly de-duplication and email failure.